

B.Tech. Degree VI Semester Supplementary Examination November 2013

ME 601 INSTRUMENTATION AND CONTROL SYSTEMS (2006 Scheme)

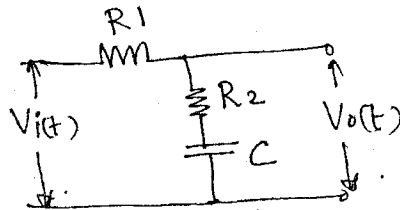
Time : 3 Hours

Maximum Marks : 100

PART A (Answer ALL questions)

(8 × 5 = 40)

- I. (a) What do you mean by zero order measurement system? Explain linear potentiometer as a zero order system.
- (b) Explain the terms static sensitivity and linearity of an instrument.
- (c) Derive an expression for gauge factor of a strain gauge.
- (d) What is a thermocouple? Explain any three laws of thermocouple.
- (e) Define transfer function. Obtain the transfer function of the given network.



- (f) Obtain the mathematical model of a typical thermal system.
- (g) What are the different frequency domain specifications?
- (h) The open loop transfer function of a unity feed back control system is given by $G(s) = \frac{K}{(s+2)(s+4)(s^2+6s+25)}$. By applying Routh - Hurwitz criterion obtain the range of 'K' for which the system to be stable.

PART B

(4 × 15 = 60)

- II. (a) Explain a typical generalized measurement system with its functional block diagram. (10)
- (b) Define the terms: (i) Accuracy (5)
(ii) Precision
(iii) Sensitivity
(iv) Hysteresis

OR

- III. What are the different types of errors in measurements and how they can be minimized? (15)

- IV. (a) Explain the working principle of an optical pyrometer with neat diagram (10)
- (b) Explain the working of hydraulic dynamometer. (5)

OR

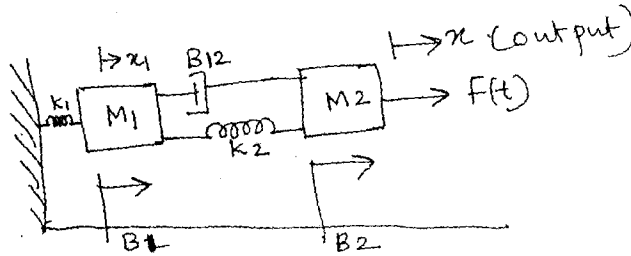
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- V. (a) Explain the working of an ORSAT's apparatus with neat sketch (10)
 (b) What do you mean by bonded and unbonded strain gauges? (5)

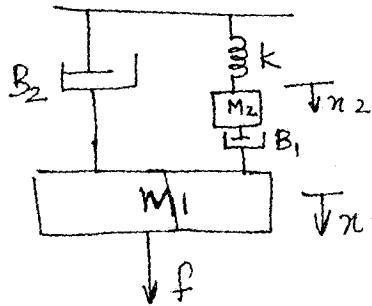
- VI. (a) What are different time domain specifications? (5)
 (b) A feed back control system has open loop transfer function $G(s) = \frac{10}{s(s+2)}$. Find (10)
 rise time, peak time, max: peak over shoot and settling time for unit step input.

OR

- VII. (a) Obtain the transfer function of the mechanical system shown below. (7)



- (b) Obtain the force voltage analogous circuit for the given mechanical system. (8)



- VIII. Sketch the bode plot for the given transfer function and determine gain margin and phase margin. (15)

$$G(s) = \frac{75(1+0.25s)}{s(s^2+16s+100)}$$

OR

- IX. A unity feed back control system has open loop transfer function (15)

$$G(s) = \frac{K}{s(s^2+4s+13)}$$

Sketch the root locus.
